



Development of Warning Methods for Planned and Unplanned Takeover Requests in a Simulated Automated Driving Vehicle

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ABSTRACT

In several circumstances, a level-three automated vehicle cannot continue driving in an automated driving mode and requests a human driver to take over. In this study, a series of experiments to examine how to provide a TOR was conducted. First, for forty-one persons, a HUD icon, earcon, seat vibration, and combinations were compared. The results indicated that the HUD icon-earcon and HUD icon-seat vibration were the most effective. Second, the combinations of A-pillar LED light and cluster icon (visual), earcon and speech message (auditory), and presence/absence of seat vibration (haptic) were compared. Thirty-six volunteers participated in the ADS failure and forty in the highway exit experiment. In the ADS failure, the combination of A-pillar LED light and seat vibration (AH) reduced the RT but can induce stress. In the highway exit, a speech message is recommended due to control stability, and the AH is not recommended due to longitudinal instability.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); HCI design and evaluation methods.

KEYWORDS

Automated vehicle, Takeover, Modality, Takeover request

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1 INTRODUCTION

According to an analysis of the reports of automated driving disengagement submitted to the California Department of Motor Vehicles (DMV) from 2015 to 2017, the most frequent cause of disengagement

was errors in perception and judgment on part of the automated driving system (49.40%), followed by driver discomfort (21.70%) [10]. By contrast, in the automated driving disengagement reports submitted between 2017 and 2019, the most frequent cause was driver discomfort (52.80%), followed by vehicle control problems (33.36%) [11]. The proportion of automated driving disengagement initiated by human operators has been increasing, which should be considered in the research and development of conditionally automated vehicles.

Although the frequency of vehicle-initiated or system-oriented disengagement has decreased, human factors must be considered, such as how well a human can resume vehicle control when the automated system does not function. In conditionally automated vehicles, vehicle control transitions between the vehicle and the driver, irrespective of whether they desired by the human driver, are inevitable. During such takeovers, the driver has the obligation and responsibility to be prepared for dynamic driving tasks (DDT) [30]. To support takeover transitions, the automated driving system (ADS) should provide an effective warning to ensure that the driver can adequately perceive a given takeover request (TOR).

In this study, we aim to derive an effective takeover warning by utilizing visual, auditory, and haptic modalities at SAE automation levels 2 and 3. We measure human reaction, vehicle control, and biosignals to compare the effectiveness of various takeover warning methods. Furthermore, this study is aimed to compare the effects of multimodal warnings in two different takeover situations (unplanned and planned ODD exit). This study offers novelty in that it gives an explanation of the overall research methods regarding TOR (literature study, warning design, and experiment). In the AutoUI community, the methods and results produced in this study may provide insight to follow-up studies on ADS development and TOR. The study is conducted in a structured manner over several stages. First, ideas pertaining to the provision of warnings based on the visual, auditory, and haptic modalities are derived by conducting a literature survey. These initial ideas related to the auditory and haptic modalities are evaluated using a driving simulator, and one design each of the visual, auditory, and haptic modalities are selected. (See 2. Single modality design: visual, auditory, and haptic modality, and [9, 19]). A draft of the design specification suitable for TOR was generated based on a literature survey, and an experiment with a simulator confirmed the effects. Then, the effects of various combinations of the aforementioned single-modality warnings are compared. (See 3. Experiments to compare the combinations of visual, auditory, and haptic modalities, and [8, 16, 18, 36]). This

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experiment verified that the multimodal warnings were more effective in TOR than the single alternatives. Finally, an experiment is conducted to compare multimodality warnings in the cases of ADS system failure and highway exit by integrating the visual, auditory, and haptic modality warnings. (See 4. Experiment and [31–33] for comparisons of multimodality warnings). Numerous studies have investigated TOR problems in automated vehicles (e.g., warning [12, 15, 24, 42], non-driving-related task [22, 35]), and we refer only to the aforementioned studies in our literature review owing to the word limit.

2 SINGLE MODALITY DESIGN: VISUAL, AUDITORY, AND HAPTIC MODALITY

To derive a takeover warning design for each of the visual, auditory, and haptic modalities, guidelines for human factors, warnings used by original equipment manufacturers (OEMs), and other prior studies related to the warnings used in partially automated vehicles were reviewed. One design was selected for visual modality based on the aforementioned methods, whereas several candidate designs were compared for the auditory modality and haptic modality based on a driving simulator experiment. The final design of each modality is summarized in Table 1 of Yun and Yang's paper [8]. As a visual warning, a round red icon with a diameter of 19 mm was projected on the HUD based on the results of previous research [19]. It was designed to blink at 3–4 Hz to express an emergency situation. As an auditory warning, an earcon combining two frequency signals of 880 Hz and 1,760 Hz was used. As a haptic warning, dynamic vibration was applied to the backrest and cushion of the driver's seat for 200 ms at frequencies of 200–250 Hz with intervals of 50 ms [19].

2.1 Visual modality warning design

Visual modality warnings can deliver information in a simple manner. However, they may be conflated with other resources that deliver vehicle information and can be easily overlooked by drivers. Yun et al. and Lee et al. [9, 19] derived three proposals for visual modality warnings: a text warning ("Take Over Immediately" displayed in a 19×86 mm rectangle) that pops up on the HUD/cluster, text warning ("Take Over Immediately" displayed in a 19×86 mm rectangle) that pops up on the interior rear-view mirror, and an icon and a text warning that pop up on the HUD/cluster. (a handle-shaped icon and the text "TAKE OVER" displayed in a circle measuring 19 mm in diameter). The color of the popup is red, text is white, and blinking cycle is 3–4 Hz. Based on these studies, we selected the icon and text warning that pop up on the HUD as the visual modality.

2.2 Auditory modality warning design

It is easy to deliver information through auditory modality warnings in the form of verbal or nonverbal signals. However, such warnings can be easily obscured by background noise inside and outside vehicles. An investigation of previous studies indicated that auditory warnings are more often provided in conjunction with visual warnings than as standalone warnings. To ensure the effectiveness of auditory warnings, Campbell et al. (2016) [21] recommended using a pitch of 500–2,500 Hz, and ISO 15006 [14] recommends

mixing a narrowband signal with a separated center frequency. Yun et al. (2018) and Lee et al. (2019) [9, 19] derived three proposals for auditory warnings. The first was Tesla firmware 8.0 that repeats two beeps with a pitch of 533 Hz and on/off durations of 200 ms/60 ms, respectively. The second was the software of Hyundai IONIQ, in which 8 beeps with a pitch of 1498 Hz and on/off durations of 90 ms/30 ms, respectively, are repeated. The last was a mixture of two beeps with pitches of 880 Hz and 1760 Hz and on/off durations of 50 ms/30 ms, respectively, where each pair of beeps is repeated four times. The volume was 15–30 dB higher than vehicle noise, but it did not exceed 90 dB. Jeon (2019) [28] compared the reaction times (RTs) of participants to earcon (nonverbal audio message), speech ("take over" in a male voice), and spearcon with (fast spoken phrase that is not recognized as speech, [38]). As a result, the RT to the earcon warning and to the speech warning were the fastest. However, because the RT to the spearcon was significantly longer than that to the earcon, the spearcon warning was found to be unsuitable for use in takeover displays. Thus, we selected earcon as our auditory warning.

2.3 Haptic modality warning design

In a level 3 or higher automated vehicle, the driver's body is likely not in contact with the steering wheel and pedals. Therefore, use of the driver's seat as a haptic interface to improve communication between the vehicle and human passengers/drivers has been discussed because the seat-human contact area is the largest [40]. An analysis of the guidelines, standards, and previous studies related to takeover, as well as OEM manuals, yielded three types of haptic warnings: a pulse-type warning with gradually increasing vibration intensity, pulse-type warning with constant vibration intensity, and wave-type warning [9, 19]. The pulse-type vibration was simultaneously applied to the entire seat back and seat pan, and the on/off durations were 200 ms/50 ms, respectively. The wave-type vibration was applied downward from the seat back to the seat pan for 2250 ms. The frequency of all oscillations was between 200 and 250 Hz. Lee et al. (2019) [19] compared the takeover RTs of 29 volunteers depending on the position of vibration in the seat (three types: seat back, seat pan, all) and the presence or absence of vibration cycles (static: continuous vibration, pulse-type: repeats in cycles of 200 ms on/50 ms off) in the KMU-DS driving simulator. The experimental results indicated no differences in the RTs of the subjects depending on the position of vibration in the seat or the presence or absence of periodic vibration. However, based on the average RT, vibration transmission through the seat back and pulse-type vibration (200 ms on/50 ms off) were found to constitute the most effective haptic warning modality.

3 EXPERIMENT TO COMPARE COMBINATIONS OF VISUAL, AUDITORY, AND HAPTIC MODALITIES

A multimodality warning is a combination of two or more warning modalities that are used simultaneously (e.g., visual, and auditory modalities) to provide a warning. In previous studies on takeover, it has been recommended that two or more modalities be used to provide takeover warnings [4, 6, 17]. In Chapter 3 of the present study, the effects of takeover are compared depending on the use of

combinations of the visual, auditory, and haptic modalities derived in Chapter 2. In most previous studies, the RTs to different types of takeover warnings have been compared [e.g. [1, 19]]. In the present study, the effects of takeover warnings are evaluated in terms of not only the driver's RT but also control stability and biosignals.

3.1 Experimental hypothesis

- H_1 : In the case of an unplanned operational design domain (ODD) exit, there are differences in human behavior, control stability, and biosignal depending on the combinations of multimodality warnings used.
- H_2 : In the case of a planned ODD exit, there are differences in human behavior, control stability, and biosignal depending on the combinations of multimodality warnings used.
- H_3 : There are differences between unplanned and planned ODD exit in terms of human behavior, control stability, and biosignal.

3.2 Experiment design and environment

3.2.1 Independent variables. The independent variables were seven combinations of the visual, auditory, and haptic modalities: visual (V), auditory (A), haptic (H), visual-auditory (VA), visual-haptic (VH), auditory-haptic (AH), and visual-auditory-haptic (VAH).

3.2.2 Dependent variables. The dependent variables in this study were as follows. The human reaction indicators were perception time (PT), human RT (HRT), and time to lane change (TTL). The control stability indicators were maximum acceleration (MA), steering angle deviation (SAD), steering reversal rate (SRR), standard deviation of lane position (SDLP), and number of gas pedal inputs (NGI). The biosignal indicators were skin conductance response onset time (SCR), average heart rate (AHR), electroencephalogram (EEG) stress, EEG attention, relative gamma power spectrum (RG), relative beta power spectrum (RB), and relative beta power spectrum (RT). Detailed descriptions of these indicators are presented in Table 1

3.2.3 Takeover scenario. Scenarios can be grouped into two types. The first type is the unplanned ODD exit scenario. During automated driving at 100 kph in the third lane of a highway, 6 s before collision with an out-of-order vehicle in front, a TOR is issued to the driver. During automated driving, the driver is supposed to be engaged in a non-driving-related task (NDRT) to make ensure that he or she does not look ahead. The driver perceives the TOR, stops the NDRT, and switches to manual driving by using one of the three controls: brake, throttle, or steering wheel. In this scenario, the driver experiences all seven types of multimodality warnings. The second scenario is the planned ODD exit scenario, in which automated driving is canceled to enter a highway junction. During automated driving at 100 kph, a TOR is issued 15 s before exiting the highway. In this scenario, the driver experiences only two types of multimodality warnings (VA and VAH) [8, 16, 18, 36].

3.2.4 Equipment. In the KMU-DS driving simulator, SCANeR studio v1.7 was used as the virtual environment engine, Facelab was used for eye tracking, and MP 160 by Biopac was used to acquire skin conductance and heart rate data.

3.2.5 Volunteers. A total of 41 volunteers were tested. Of these, 25 were men, and 16 were women. Their average age was 26.2 years, and their average driving experience was 3.7 years.

3.2.6 Procedure. The volunteers wore the biosignal acquisition equipment after filling out the Institutional Review Board (IRB) consent form. Then, they practiced driving to learn how to take over control and operate the simulator. After completing the practice, they took the main driving test. They experienced all nine scenarios in a random order. When the volunteers completed one scenario, a survey was conducted to evaluate the warnings they experienced. After confirming the data, the volunteers removed the biometric equipment, and the experiment was ended. The experiment duration was 150 min.

3.3 Result

3.3.1 Comparison of modality combinations in unplanned ODD exits (H_1). The comparison results of multimodality warning combinations for the unplanned ODD exit scenario are shown in Table 2 (unplanned ODD exit column). The shorter the human behavior index, the smaller is the control stability index, higher is the biosignal index, and higher is the ranking. In summary, the visual-only modality performed the worst on all indices compared to all other combinations. Combined modalities with either the auditory or haptic modality as one component were more effective than other combinations.

3.3.2 Comparison of modality combinations in planned ODD exits (H_2). Table 2 presents a comparison of the multimodality warning combinations for the planned ODD exit scenario (planned ODD exit column). VAH led to a shorter reaction time (HRT) than VA, but a longer TTL. No significant difference was observed between VAH and VA in terms of control stability and biosignal.

3.3.3 Comparison of unplanned and planned ODD exits (H_3). In the planned ODD exit scenario, only the VA and VAH warnings were provided. When the VA and VAH warnings were used, the differences between the unplanned ODD exit and planned ODD exit were as follows. In case of the VA warning, the TTL of the unplanned ODD exit was significantly shorter than that of the planned ODD exit ($p < 0.001$). There was a small but significant difference in SDLP ($p < 0.001$). In case of the VAH warning, the HRT and TTL of the unplanned ODD exit scenario were significantly shorter than those of the planned ODD exit scenario ($p = 0.006$; $p < 0.001$). Moreover, SDLP and SRR were significantly smaller ($p < 0.001$, $p < 0.001$). However, there were no significant differences between SCR and AHR [8].

3.3.4 Interpretation of the results. The visual modality alone should not be used in a takeover situation because it induces a slow reaction from the driver [36] and reduces control stability [16]. Moreover, it induces low stress and attention levels in the driver [18]. Multimodality warnings should be used to provide takeover warnings. VAH yielded good takeover performances in terms of human behavior, control stability, and biosignal. However, it can increase driver workload [8, 36]. In addition, in terms of control stability, VAH or VH led to high lateral control stability. AH is effective for longitudinal control stability [16]. In terms of EEG, VAH was the

Table 1: Dependent variables in multimodality experiment.

Category	DV [unit]	Definition	Meaning	Reference
Human behavior	Perception time (PT) [s]	Time between recognition of TOR and moving eyes to check the situation	Evaluate whether the driver intuitively understood the TOR situation.	[36]
	Human reaction time to TOR (HRT) [s]	Time between TOR and resumption of manual driving	Evaluate whether the driver immediately responds to the TOR situation.	[8, 36]
	Time to lane change (TTL) [s]	Time between TOR and lane change	Evaluate whether the driver immediately responded to the TOR situation.	[8, 36]
Control stability	Maximum acceleration (MA) [m/s ²]	Maximum acceleration of vehicle after takeover	The smaller the MA after manual driving, the more stable is the takeover.	[2, 16]
	Steering wheel angle deviation (SAD) [degree]	Absolute value of difference between the maximum and minimum steering angles	The greater the steering wheel angle deviation, the lower is the driving stability and the greater is the cognitive load.	[3, 16, 37]
	Steering reversal rate (SRR) [count/s]	Measured as the frequency of steering reversal when the steering angle difference exceeds 3°	Used to evaluate driver distraction. The more difficult the driving task, the higher is the SRR.	[7, 16, 20, 26]
	Standard deviation of lane position (SDLP) [m]	Standard deviation of lateral position of the vehicle from the center of the lane	As an evaluation of lane keeping, the higher the SDLP, the lower is the driving stability.	[5, 8, 16, 23, 25]
	Number of gas pedal inputs (NGI) [count/10 s]	Number of times the accelerator pedal is pressed after takeover	Driver's effort to control the vehicle in the longitudinal direction. The larger its value, the more unstable is the longitudinal control.	[16, 26]
Biosignal	Skin conductance response onset time (SCR) [s]	Immediately after a TOR, the time from the start of increase in skin conductance to the point where it reaches its maximum amplitude	The greater the stress, the faster is the increase in SCR.	[8, 29]
	Average heart rate (AHR) [bpm]	Average heart rate of drivers after TOR	The higher the AHR, the more stressed is the driver.	[8, 13]
	EEG stress [-]	Ratio of beta power (13–30 Hz) to alpha power (8–12 Hz). Beta/alpha as an index indicating stress	The higher the value, the more stressed is the driver.	[18, 39]
	EEG attention [-]	Ratio of theta power activated during sleep (4–8 Hz) to SMR activated during concentration (13–15 Hz)	A variable used as a degree of concentration on driving. The higher the value, the greater is the concentration on driving.	[18]
	Relative gamma power spectrum (RG) [-]	The ratio of gamma power (30–50 Hz) activated during high concentration to 0–50 Hz	The higher the value, the greater is the concentration on driving.	[18]
	Relative beta power spectrum (RB) [-]	Ratio of beta power (13–30 Hz) activated during concentration to 0–50 Hz	The higher the value, the greater is the concentration on driving.	[18]
	Relative theta power spectrum (RT) [-]	Ratio of theta power (4–8 Hz) activated during sleepiness or meditation to 0–50 Hz	A higher value indicates a relatively sleepy state.	[18]

Table 2: Comparison results of multimodality warning combinations.

Category	DV [unit]	Result	Ranking(Unplanned ODD exit)	Ranking(Planned ODD exit)
Human behavior	Perception time (PT) [s]	V had a slower PT than other warning combinations.	All except V > V	-
	Human reaction time to TOR (HRT) [s]	VH and VAH had shorter HRT than other warnings.	VH, VAH > H > VA, AH, A > V	VAH > VA
	Time to lane change (TTL) [s]	AH, and VAH led to shorter times to lane change than other warnings.	AH, VAH > A, VH > VA, H > V	VA > VAH
Control stability	Maximum acceleration (MA) [m/s ²]	A and H had the smallest MA.	A, H > VA, AH, VH > VAH > V	No significant difference
	Steering angle deviation (SAD) [degree]	AH, VA, and A led to the least SAD.	AH, VA, A > H, VH > VAH > V	No significant difference
	Steering reversal rate (SRR) [count/s]	VA led to the lowest SRR compared to the other warning combinations.	VA > A, AH, H, VH > VAH > V	No significant difference
	Standard deviation of lane position (SDLP) [m]	SDLP was the least when VAH, VH, and A were provided.	VAH, VH, A > AH, VA > H > V	No significant difference
	Number of gas pedal inputs (NGI) [count/10 s]	NGI was small when AH was provided.	AH > VAH, VH, A, VA > H > V	No significant difference
Biosignal	Skin conductance response time (SCR) [s]	The SCR was the least when V was provided.	All except V > V	No significant difference
	Average heart rate (AHR) [bpm]	No differences in AHR across warning combinations.	No significant difference	No significant difference
	EEG stress [-]	A led to higher EEG stress than the other warning combinations.	A > VAH, H > VA, AH, VH > V	-
	EEG attention [-]	VAH, H, VH, and AH led to higher EEG attention than the other warning combinations.	VAH, H, VH, AH > VA, A > V	-
	Relative gamma power spectrum (RG) [-]	A, and VAH led to higher RG than the other warning combinations.	A, VAH > V, H, VA, VH, AH	-
	Relative beta power spectrum (RB) [-]	VAH led to higher RB than the other warning combinations.	VAH > VA, A, H, AH > VH, V	-
	Relative theta power spectrum (RT) [-]	VA led to higher RT than the other warning combinations.	VA > V > VH, VAH, AH, A > H	-

most effective for providing TORs because it made the driver attentive [18]. In sum, it is recommended that to provide a TOR, the visual modality should be combined with the auditory or haptic modality [8, 17]. Use of the visual modality alone in a takeover situation adversely affects the takeover, but it can express the driving situation in detail. Volunteers reacted more slowly to the planned ODD exit situation than to the unplanned ODD exit situation, even if the same modality was used to provide the TOR [8]. Each method has its own advantages. For example, a haptic warning may induce a more immediate takeover, whereas an auditory warning may induce a more stable takeover.

4 EXPERIMENT TO COMPARE VARIOUS MULTIMODAL WARNING METHODS

Based on the findings in Chapter 3, we recommend using the visual and auditory or haptic modalities together. Considering these results and by diversifying the visual and auditory design variables, the effects of multimodal warning methods on TOR are compared

in this section. The warning designs of each modality are presented in Table 1 in Hong and Yang's paper [31].

4.1 Experimental hypothesis

- H₁: PT, HRT, MA, SAD, and SCR amplitude differ depending on various combinations of visual warnings (A-pillar LED lights + cluster icon vs. cluster icon only).
- H₂: PT, HRT, MA, SAD, and SCR amplitude differ depending on the types of auditory warnings (earcon vs. speech message (same as sentence-based speech)).
- H₃: PT, HRT, MA, SAD, and SCR amplitude differ depending on the presence or absence of vibration in the seat.
- H₄: The interaction effect depends on the design method of the visual, auditory, and haptic warnings.

4.2 Multimodal warning design

4.2.1 Visual modality. An LED light that can be turned on in the A-pillar was used to provide a visual warning (A). The purpose of using a light on the A-pillar was to investigate the effect of the

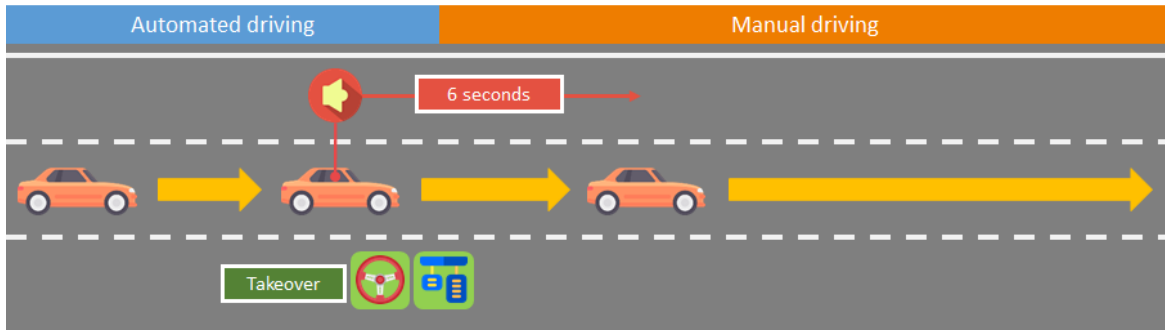


Figure 1: ADS system failure scenario.

light in terms of stimulating the driver's peripheral vision during the takeover compared to their focal vision. A moving cue was activated on the LED strip installed on the A-pillar when the TOR was provided. The illuminance and color of the moving cue were 1,500 lx (inside the simulator: 2.10 lx) and red (R:255, G:0, B:0), respectively. The length and width of the LED strip were 500 mm and 10 mm, respectively. The LED number was 30EA, LED lens diameter was 5 mm, and frequency was 2.86 Hz. In addition, an icon was projected on the cluster (C). During driving in the automated mode, a green icon was provided. During a TOR, the green icon was changed to a red icon with the phrase "Take over." This icon was developed by Stanford University and Korea Automobile Research Institute. (Photo 1 in [41]).

4.2.2 Auditory modality. An earcon (E) and a speech message (S) were used as auditory modalities. The earcon tone was created by mixing the tones of Tesla firmware 8.0 and Hyundai IONIQ, and its pitch was 880 Hz + 1760 Hz (using the same auditory warning as in 3.2) [28]. The speech message was "Drive yourself" in Korean, and the pitch was 217 Hz [34]. Both warnings were provided at a volume of 70 dB, and they were repeated until the DDT fallback was completed.

4.2.3 Haptic modality. The presence (H) and absence (N) of haptic seat was compared in the multimodality warning method experiment. A vibration seat was installed in the seat back and seat pan. Vibration with a frequency of 28–34.5 Hz, magnitude of 2g, and duration of 200 ms/50 ms on/off, respectively, was repeated after TOR (by using the same haptic warning as that in 3.2).

4.3 Experiment Design and Simulation Environment

4.3.1 Experiment design. Each participant experienced all eight types of multimodality warnings. Combinations of "A-pillar LED light and cluster icon (A) vs. cluster icon (C) for visual modality," "earcon (E) vs. speech messages (S) for auditory modality," "haptic seat (H) vs. no haptic seat (N) for haptic modality" resulted in the following eight cases: AEH, AEN, ASH, ASN, CEH, CEN, CSH, and CSN. To minimize the participants' learning effect that may occur when using the within-subject design, a Latin square design combining TOR and time point was used. Two scenarios without

TOR were added; it was not explained to the volunteers in advance whether a TOR was issued.

4.3.2 Takeover scenarios. In the first scenario, automated driving was disengaged due to a malfunction of the ADS system during automated driving at 85 kph as an example of unplanned ODD exit (Figure 1). After the TOR, the driver was expected to take over control and switch to manual driving by using one of the following three controls: brake, throttle, or steering wheel. After the driver controlled the vehicle for more than 30 s and their skin conductance level returned to the baseline, the scenario ended.

In the second scenario, automated driving was disengaged due to highway exit as an example of planned ODD exit. During automated driving at 85 kph, a TOR was provided 15 s before the vehicle's entry into the branch. The driver was expected to take over control and switch to manual driving by using one of the three controls: brake, throttle, or steering wheel. After that, once the car exited the highway, the scenario ended (Figure 2).

4.3.3 Experimental environment. The experiment was conducted using a driving simulator. In the cabin, a 5.1-channel speaker was installed to provide auditory warnings, and nine vibration motors were installed in the driver's seat to provide haptic warnings. As the simulation software, SCANer studio ver. 1.8 (AV Simulation) was used. Biosignal data, namely skin conductance data of the volunteers, were obtained using a Biopac MP160. Driver gaze data were obtained using a Smart Eye Pro 3 Cam system (Figure 3).

4.3.4 Volunteers. Thirty-six people participated in the unplanned ODD exit scenario, of which 22 were male and 14 were female. The average age of the volunteers was 40.9 years, and their driving experience was 4.5 years. Forty people, that is, 20 men and 20 women, participated in the planned ODD exit scenario. Their average age was 40.2 years, and their driving experience was 12.3 years. This study followed the protocol approved by Kookmin University Institutional Review Board (KMU-201803-HR-174).

4.3.5 Independent variables. The independent variables were 1) visual modality (A and C), 2) auditory modality (E and S), and 3) haptic modality (H and N).

4.3.6 Dependent variables. The dependent variables were PT, HRT, MA, SAD, and SCR amplitude. The definitions of PT, RT, MA, and SAD were the same as those in 3.3 (Table 1). SCR amplitude refers to

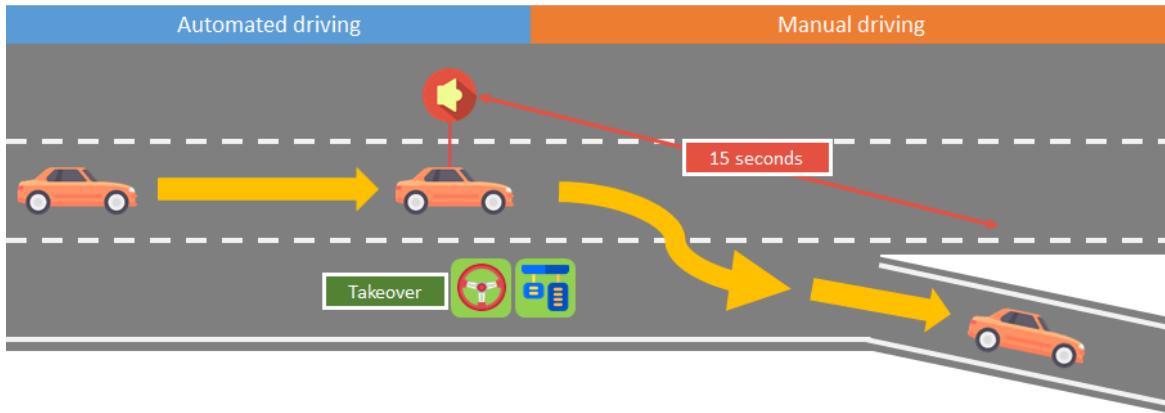


Figure 2: highway exit scenario.



Figure 3: Experimental environment

the difference between the baseline and maximum conductance. In this study, it was defined as the difference between the baseline of skin conductance and the peak value of conductance immediately after the provision of a TOR [26].

4.4 Result

A three-way repeated measure ANOVA was performed to analyze the main effects and interaction effects of the visual, auditory, and haptic warnings.

4.4.1 Main effect of visual modality warnings (H_1). When automated driving was disengaged due to ADS system failure, the average HRT of the A-pillar LED light was 2.27 s, which was significantly faster than the cluster-only average HRT of 2.51 s ($p = 0.022$). The average SAD of the A-pillar LED light was 8.24° , which was significantly greater than the cluster-only average SAD of 6.44° ($p = 0.024$) [5]. In case of the planned ODD exit due to highway exit, the average HRT to the A-pillar LED light was 3.05 s, which was significantly faster than the average HRT of 3.28 s to warnings without the A-pillar LED light ($p = 0.006$) (Table 3).

4.4.2 Main effect of auditory modality warnings (H_2). When automated driving was disengaged due to ADS system failure, the average SAD due to an earcon was 6.66° , which was significantly lower than the average SAD of 8.02° due to speech messages ($p = 0.044$). By contrast, when automated driving was automated due to a planned exit, the average MA due to the earcon was 3.34 m/s^2 ,

which was significantly higher than the average MA of 3.05 m/s^2 due to the speech message ($p = 0.028$). The average SAD due to the earcon was 43.18° , which was significantly greater than the average SAD of 38.06° due to the speech message ($p = 0.037$) (Table 3).

4.4.3 Main effect of haptic modality warnings (H_3). When automated driving was disengaged due to ADS system failure, the average HRT to the haptic seat was 2.21 s, which was significantly faster than the average HRT of 2.58 s with no haptic warning ($p < 0.001$). The average SCR amplitude of the haptic seat was $3.21 \mu\text{s}$, which was significantly higher than the average SCR amplitude of $2.53 \mu\text{s}$ with no haptic warning ($p = 0.001$). In the planned ODD exit scenario, the average PT with the haptic seat was 0.97 s, which was significantly faster than the average PT of 1.14 s with no haptic warning ($p = 0.025$). The average HRT of the haptic seat was 3.02 s, which was significantly higher than the average HRT of 3.31 s with no haptic warning ($p = 0.003$) (Table 3).

4.4.4 Interaction effect of visual, auditory, and haptic modality warnings (H_4). When automated-driving was disengaged due to ADS system failure, the average SAD of the A-pillar-earcon combination was 6.01° , which was lower than the average SAD of 10.46° with the A-pillar-speech message combination. The average SAD with the cluster-earcon combination was 7.31° , which was greater than the average SAD of 5.58° with the cluster-speech message combination ($p < 0.001$). The average SCR amplitude with the A-pillar-earcon combination was $2.86 \mu\text{s}$, which was lower than the average SCR

Table 3: Experimental results obtained using various warning modalities.

Modality	Takeover scenario	PT	HRT	MA	SAD	SCR amplitude
Visual (H ₁)	ADS system failure	-	A > C	-	A > C	-
	Highway exit	-	A > C	-	-	-
Auditory (H ₂)	ADS system failure	-	-	-	E > S	-
	Highway exit	-	-	S > E	S > E	-
Haptic (H ₃)	ADS system failure	-	H > N	-	-	H > N
	Highway exit	H > N	H > N	-	-	-
V x A (H ₄)	ADS system failure	-	-	-	AE > AS, CS > CE	AS > AE, CE > CS
	Highway exit	-	-	-	-	-
V x H (H ₄)	ADS system failure	-	-	-	-	-
	Highway exit	-	-	AN > AH, CH > CN	-	-
A x H (H ₄)	ADS system failure	-	-	-	-	-
	Highway exit	-	-	-	-	-
V x A x H (H ₄)	ADS system failure	-	-	-	-	-
	Highway exit	-	-	-	-	-

A: A-pillar LED light, C: cluster-only, E: earcon, S: speech message, H: haptic, N: No-haptic, -: no significant difference

amplitude of $3.06 \mu s$ with the A-pillar-speech message combination. The mean SCR amplitude with the cluster-earcon (CE) combination was $3.10 \mu s$, which was higher than the mean SCR amplitude of $2.44 \mu s$ with the cluster-speech message combination ($p = 0.019$). In the planned ODD exit scenario, the average MA with the A-pillar-haptic modality combination was $3.21 m/s^2$, which was higher than the average MA of $2.99 m/s^2$ with the A-pillar-no-haptic combination. The mean MA of the cluster-no-haptic combination was $3.46 m/s^2$, which was higher than the value of $3.12 m/s^2$ with the cluster-haptic combination ($p = 0.024$) (Table 3).

4.4.5 Interpretation of the results. Two scenarios, ADS system failure and highway exit, were used to represent an unpredictable situation (unplanned ODD exit) and a planned situation (planned ODD exit), respectively, for disengaging automated driving. In case of the highway exit, the haptic seat warning induced quick reactions from the drivers. Regardless of the takeover situation, the A-pillar LED light and haptic seat warnings induced quick reactions from the drivers. In terms of vehicle indicators, the effects of the warnings differed depending on the takeover situation. This difference was ascribed to the fact that the driving tasks to be conducted by the volunteers after the takeover were different. In the ADS system failure case, there was a significant difference in the lateral direction indicator (SAD) depending on the TOR modality. The speech message warning led to higher lateral driving safety than the earcon warning. The combination of A-pillar LED light and earcon warning and the speech message warning in the absence of A-pillar LED light warning induced stable lateral control. By contrast, in the highway exit situation, there was a significant difference in the longitudinal index (MA) depending on the TOR method. The speech message warning yielded higher longitudinal

driving safety than the earcon warning. Moreover, when the A-pillar LED light warning and the haptic seat warning were not used or when the haptic seat warning was used without the A-pillar LED light warning, the longitudinal driving safety was high. In terms of biosignal, in the ADS system failure situation, the haptic seat warning induced greater driver stress. Moreover, the combination of A-pillar LED light and speech message warnings or the earcon warning alone without the A-pillar LED light warning induced greater stress in the drivers. An advantage of this was that the vehicle clearly delivered the warning content to the driver (and induced a quick reaction). However, the disadvantage was driver discomfort. When designing a TOR for an automated driving system, a tradeoff between eliciting quick reactions from the driver and sensibility is necessary.

5 CONCLUSION

In this paper, we summarized a four-year long study on the development of warning methods for planned and unplanned TORs in a simulated automated driving. We first started with a study on a single modality—initial example designs of the visual, auditory, and haptic modalities were proposed based on an extensive literature review and a few simulator experiments (Chapter 2, [9, 19]). These initial designs were used to test and compare the effects of TORs provided as combinations of various modalities (Chapter 3, [8, 16, 18, 36]) with the data of 41 volunteers (amounting to 509 GB). The results suggested that combinations of visual and auditory or haptic modality warnings should be used (multimodality warning), whereas visual only warnings should be avoided. In the next experiment, additional designs for visual and auditory modalities were investigated for both planned and unplanned TOR scenarios (Chapter 4, [31–34]) with the data of 76 participants (36 for unplanned and 40 for planned) (amounting to 908.01 GB). When using the A-pillar

LED light warning, it is recommended to use an earcon as an auditory warning in combination. Doing so can increase lateral driving stability than using speech message and reduce driver stress. The use of speech message warnings in combination with A-pillar LED lights is not recommended. Speech message warnings alone can be used in less urgent situations because the level of urgency conveyed by speech messages is relatively lower than that conveyed by earcon warnings. In the case of emergent situations where fast reactions are necessary, the use of a haptic seat warning is recommended, but it may stress the driver. During a planned takeover (when a lane change is required), speech message warnings should be used to enhance longitudinal driving stability. Moreover, the A-pillar LED light and the haptic seat warning should not be used together. In summary, when designing TORs, it is necessary to consider both the urgency of the takeover situation and driver acceptance.

6 LIMITATIONS AND FUTURE WORK

This experiment was conducted considering limited takeover scenarios. Therefore, future studies should verify the warning effects in more diverse takeover scenarios by selecting a few warnings with excellent performance. In addition, a follow-up study is needed to verify whether the effects of the warnings persist for drivers who have been repeatedly exposed to takeover experiences. During the experiment, a few volunteers found the haptic warning unpleasant. Because emotions affect an individual's driving performance [27], further research on negative emotions or feelings is needed. Research on the subdivision of TORs depending on the degree of risk of a takeover situation is expected to be conducted in the future. In addition, it is necessary to develop more detailed TOR specifications that can be applied to mass-produced systems.

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REFERENCES

- [1] Alexander Eriksson, Sebastiaan M. Petermeijer, Markus Zimmermann, Joost C. F. de Winter, Klaus J. Bengler and Neville A. Stanton. 2018. Rolling out the red (and green) carpet: supporting driver decision making in automation-to-manual transitions. *IEEE Transactions on Human-Machine Systems*. 49 (February 2018), 20-31. 10.1109/THMS.2018.2883862
- [2] Christian Gold, Daniel Damböck, Lutz Lorenz, Klaus Bengler. 2013. "Take over!" How long does it take to get the driver back into the loop?. In *Proceedings of the human factors and ergonomics society 57th annual meeting*. Sage CA: Los Angeles, CA: Sage Publications. 57, 1, 1938-1942. <https://doi.org/10.1177/1541931213571433>
- [3] Daniel V. McGehee, John D. Lee, Matthew Rizzo, Jeffrey Dawson and Kirk Bateman. 2004. Quantitative analysis of steering adaptation on a high performance fixed-base driving simulator. *Transportation Research Part F: Traffic Psychology and Behaviour*, 7, 3 (May 2007), 181-196. <https://doi.org/10.1016/j.trf.2004.08.001>
- [4] Dongyeon Yu, Chanhoo Park, Hoseung Choi, Donggyu Kim and Sung-Ho Hwang. 2021. Takeover Safety Analysis with Driver Monitoring Systems and Driver-Vehicle Interfaces in Highly Automated Vehicles. *Applied Sciences*, 11, 15 (July 2021), 6685. <https://doi.org/10.3390/app11156685>
- [5] Frederik Naujoks, Christoph Mai and Alexandra Neukum. 2014. The effect of urgency of take-over requests during highly automated driving under distraction conditions. *Advances in human aspects of transportation*, 7(Part I), 431-438.
- [6] Frederik Naujoks, Katharina Wiedemann, Nadja Schömig, Sebastian Hergeth and Andreas Keinath. 2019. Towards guidelines and verification methods for automated vehicle HMI. *Transportation research part F: traffic psychology and behaviour*, 60 (January 2019), 121-136. <https://doi.org/10.1016/j.trf.2018.10.012>
- [7] Gustav Markkula and Johan Engström. 2006. A steering wheel reversal rate metric for assessing effects of visual and cognitive secondary task load. In *Proceedings of the 13th ITS World Congress*, The University of Leeds, London, UK.
- [8] Hanna Yun and Ji Hyun Yang. 2020. Multimodal warning design for take-over request in conditionally automated driving. *European transport research review*, 12, 1 (May 2020), 1-11. <https://doi.org/10.1186/s12544-020-00427-5>
- [9] Hanna Yun, Ji Won Lee, Hee Dong Yang and Ji Hyun Yang. 2018. Experimental design for multi-modal take-over request for automated driving. In *International Conference on Human-Computer Interaction (HCI International 2018)*. 418-425. 10.1007/978-3-319-92285-0_57
- [10] Hanna Yun, Sung Lae Kim, Ji Won Lee and Ji Hyun Yang. 2018. Analysis of Cause of Disengagement Based on U.S. California DMV Autonomous Driving Disengagement Report. *Transaction of the Korean Society of Automotive Engineers*, 26, 4 (Jun 2018), 464-475. <https://doi.org/10.7467/KSAE.2018.26.4.464>
- [11] Haozhong Zuo and Ji Hyun Yang. 2021. Analysis of Cause of Disengagement Based on U.S. California DMV Autonomous Driving Disengagement Report (2017-2019). In *Proceeding of the Korean Society of Automotive Engineers*. (November 2021), 423.
- [12] Harsh Sanghavi, Yiqi Zhang and Myoungsoon Jeon. 2020. Effects of anger and display urgency on takeover performance in semi-automated vehicles. In *Proceedings of the 12th International Conference on Automotive User Interfaces and Interactive Vehicular Applications (AutomotiveUI '20)*, 48-56. <https://doi.org/10.1145/3409120.3410664>
- [13] Hyung Jun Kim and Ji Hyun Yang. 2017. Takeover requests in simulated partially autonomous vehicles considering human factors. *IEEE Transactions on Human-Machine Systems*. 47, 5 (March 2017), 735-740. <https://doi.org/10.1109/THMS.2017.2674998>
- [14] ISO 15006. 2011. Road vehicles - Ergonomic aspects of transport information and control system - specifications for in-vehicle auditory presentation.
- [15] J.J. Scott and Robert Gray. 2008. A comparison of tactile, visual, and auditory warnings for rear-end collision prevention in simulated driving. *Human factors*. 50, 2 (April 2008), 264-275. <https://doi.org/10.1518/001872008X250674>
- [16] Jae Won Kim and Ji Hyun Yang. 2020. Understanding metrics of vehicle control take-over requests in simulated automated vehicles. *International journal of automotive technology*, 21, 3 (February 2020), 757-770. <https://doi.org/10.1007/s12239-020-0074-z>
- [17] Ji Hyun Yang, Seul Chan Lee, Chihab Nadri, Jaewon Kim, Jaekon Shin and Myoungsoon Jeon. Multimodal displays for takeover requests. In *User Experience Design in the Era of Automated Driving*. Springer, 2022, 397-424. 10.1007/978-3-030-77726-5_15
- [18] Jiwon Lee and Ji Hyun Yang. 2020. Analysis of driver's EEG given take-over alarm in SAE level 3 automated driving in a simulated environment. *International journal of automotive technology*, 21, 3 (February 2020), 719-728. <https://doi.org/10.1007/s12239-020-0070-3>
- [19] Jiwon Lee, Hanna Yun, Jaewon Kim, Sujin Baek, Hyunseo Han, S. Maryam Fakhr Hosseini, Eric Vasey, Okkeun Lee, Myoungsoon Jeon and Ji Hyun Lee. 2019. Design of Single-modal Take-over Request in SAE Level 2 & 3 Automated Vehicle. *Transaction of the Korean Society of Automotive Engineers*, 27, 3 (March 2019), 171-183. <http://doi.org/10.7467/KSAE.2019.27.3.171>
- [20] Joakim Östlund, Lena Nilsson, Jan Törnros and Åsa Forsman. 2006. Effects of cognitive and visual load in real and simulated driving. *Swedish National Road and Transport Research Institute*.
- [21] John L. Campbell, James L. Brown, Justin S. Graving, Christian M. Richard, Monica G. Lichty, Thomas Sanquist, L. Paige Bacon, Robert Woods, Hong Li, Diane N. Williams and Justin F. Morgan. 2016. Human factors design guidance for driver-vehicle interfaces. Report No. DOT HS, 812 360. National Highway Traffic Safety Administration.
- [22] Jonas Radlmayr, Christian Gold, Lutz Lorenz, Mehdi Farid and Klaus Bengler. 2014. How traffic situations and non-driving related tasks affect the take-over quality in highly automated driving. In *Proceedings of the human factors and ergonomics society 58th annual meeting*. 2063-2067. <https://doi.org/10.1177/1541931214581434>
- [23] Joris C Verster and Thomas Roth. 2011. Standard operation procedures for conducting the on-the-road driving test, and measurement of the standard deviation of lateral position (SDLP). *International journal of general medicine*, 4 (May 2011), 359-371. 10.2147/IJGM.S19639

- [24] Jusuf Çapalar and Cristina Olaverri-Monreal. 2017. Hypovigilance in limited self-driving automation: Peripheral visual stimulus for a balanced level of automation and cognitive workload. In 2017 IEEE 20th International Conference on Intelligent Transportation Systems (ITSC). 27-31.10.1109/ITSC.2017.8317925
- [25] Karel A. Brookhuis, Gerbrand de Vries and Dick De Waard. 1991. The effects of mobile telephoning on driving performance. *Accident Analysis & Prevention*, 23, 4 (August 1991), 309-316. [https://doi.org/10.1016/0001-4575\(91\)90008-S](https://doi.org/10.1016/0001-4575(91)90008-S)
- [26] M. Mulder, S. Kitazaki, S. Hijikata, Max. Mulder, M.M. van Paassen and E.R. Boer. 2004. Reaction-time task during car-following with an active gas pedal. In 2004 IEEE International Conference on Systems, Man and Cybernetics (IEEE Cat. No. 04CH37583). 3, 2465-2470. 10.1109/ICSMC.2004.1400700
- [27] Myounghoon Jeon. 2017. Emotions and affect in human factors and human-computer interaction: Taxonomy, theories, approaches, and methods. In *Emotions and affect in human factors and human-computer interaction*. (April 2017), 3-26. <https://doi.org/10.1016/B978-0-12-801851-4.00001-X>
- [28] Myounghoon Jeon. 2019. Multimodal displays for take-over in level 3 automated vehicles while playing a game. In *Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems*. (May 2019). 1-6. <https://doi.org/10.1145/3290607.3313056>
- [29] Reiner Nikula. 1991. Psychological correlates of nonspecific skin conductance responses. *Psychophysiology*, 28, 1 (January 1991), 86-90. <https://doi.org/10.1111/j.1469-8986.1991.tb03392.x>
- [30] SAE On-Road Automated Vehicle Standards Committee. 2021. Taxonomy and definitions for terms related to driving automation systems for on-road motor vehicles. Warrendale: SAE International. https://www.sae.org/standards/content/j3016_202104/
- [31] Sara Hong and Ji Hyun Yang. 2022. Effect of multimodal takeover request issued through A-pillar LED light, earcon, speech message, and haptic seat in conditionally automated driving (UNDER REVIEW).
- [32] Sara Hong, Ho Yung Shim and Ji Hyun Yang. 2020. Simulator study on take-over request under ADS system failure and highway exit situation. *KSAE Annual Conference Proceedings*, 516.
- [33] Sara Hong, Ho Yung Shim and Ji Hyun Yang. 2021. Simulator study and guidelines for take-over request in partially automated vehicle. In *Proceedings of 2021 Spring Conference of ESK*, 79-80.
- [34] Sara Hong, Jae Won Kim, Su Jin Baek and Ji Hyun Yang. 2020. Simulator Study on Take-over Request in Partially Automated Vehicle Using Auditory and Haptic Modality. *Transaction of the Korean Society of Automotive Engineers*, 28, 6 (June 2020), 401-411. <https://doi.org/10.7467/KSAE.20120.28.6.401>
- [35] Sol Hee Yoon, Young Woo Kim and Yong Gu Ji. 2019. The effects of takeover request modalities on highly automated car control transitions. *Accident Analysis & Prevention*, 123 (February 2019), 150-158. <https://doi.org/10.1016/j.aap.2018.11.018>
- [36] Su Jin Baek, Hanna Yun, and Ji Hyun Yang. 2019. How do humans respond when automated vehicles request an immediate vehicle control take-over?. In *Proceedings of the 11th International Conference on Automotive User Interfaces and Interactive Vehicular Applications: Adjunct Proceedings (AutomotiveUI '19)*. 341-345. <https://doi.org/10.1145/3349263.3351496>
- [37] Tania Dukic, Lars Hanson and Torbjörn Falkmer. 2006. Effect of drivers' age and push button locations on visual time off road, steering wheel deviation and safety perception. *Ergonomics*, 49, 1 (Feb 2007), 78-92. <https://doi.org/10.1080/00207540500422320>
- [38] Tilman Dingler, Jeffrey Lindsay and Bruce N Walker. 2008. Learnability of sound cues for environmental features: Auditory icons, earcons, spearcons, and speech. In *proceedings of the 14th International Conference on Auditory Display (ICAD2008)*, Paris, France, 24-27. <http://hdl.handle.net/1853/49940>
- [39] Yannick Forster, Frederik Naujoks, Alexandra Neukum and LynnHuestegge. 2017. Driver compliance to take-over requests with different auditory outputs in conditional automation. *Accident Analysis & Prevention*, 109 (December 2017), 18-28. <https://doi.org/10.1016/j.aap.2017.09.019>
- [40] Yong Gu Ji, Kwangil Lee and Wonil Hwang. 2011. Haptic perceptions in the vehicle seat. *Human Factors and Ergonomics in Manufacturing & Service Industries*, 21, 3 (May/June 2011), 305-325. <https://doi.org/10.1002/hfm.20235>
- [41] Youngseok Lee, Yeppeun Lee and Changhyun Jeong. 2019. Development of evaluation environment for driver-vehicle interaction research in Level 3 automated vehicle based on a real road. *KSAE Annual Conference Proceedings*, 673.
- [42] Yutong Li, Chen Lv and Junliang Xue. 2019. A novel predictive haptic control interface for automation-to-human takeover of automated vehicles. In *2019 IEEE Intelligent Vehicles Symposium (IV)*. (June 2019), 994-999. 10.1109/IVS.2019.8814252